

Differential equations

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1 Differential equations classification

2 Ordinary differential equations (ODE)

2.1 Separation of variables

2.2 Homogeneous ODE 1st degree

An homogeneous ODE can be expressed like:

$$y' + M(x)y = 0$$

It is always useful to use Leibniz's notation for differential equations even though we might always see the statements written using either Newton or Lagrange's notation, thus the general way to write an homogeneous ODE turns into:

$$\frac{dy}{dx} + M(x)y = 0$$

This case is super simple, because we can separate variables here.

$$\frac{dy}{dx} = -M(x)y \implies \frac{1}{y} \frac{dy}{dx} = -M(x) \implies \frac{1}{y} dy = -M(x) dx$$

Now that we have all y terms in one side and all x terms in the other, we can integrate both side:

$$\int \frac{1}{y} dy = \int -M(x) dx \implies \ln(y) + C = - \int M(x) dx$$

We are interested in the value y , not in its logarithm, hence we must subtract both sides by the integration constant and use the exponential function to cancel it.

$$y = \exp \left(- \int M(x) dx - C \right), \quad \exp(x) = e^x$$

We can use here the properties of the exponential function to simplify the whole expression.

$$y = \exp \left(- \int M(x) dx \right) e^{-C}$$

As e and $-C$ are two constants and C is absolutely arbitrary, we can rename it to just have an arbitrary value per se instead of a complex expression-

$$y = \lambda \exp \left(- \int M(x) dx \right), \quad \lambda \in \mathbb{R}$$

2.3 Non-homogeneous ODE 1st degree

A non-homogeneous ODE of 1st degree looks like:

$$y' + M(x)y = N(x)$$

As always, we switch first to the Leibniz's notation.

$$\frac{dy}{dx} + M(x)y = N(x)$$

Before solving this ODE, let us compute the following formula:

$$\frac{d}{dx} \exp \left(\int M(x) dx \right) y$$

We have the derivative of the multiplication of two functions, so we will be using the formula for it here.

$$\frac{d}{dx} \exp \left(\int M(x) dx \right) y = M(x) \exp \left(\int M(x) dx \right) y + \exp \left(\int M(x) dx \right) y'$$

We see that in the result we have in both terms the exponent result which means we can factor it out.

$$\begin{aligned} &= \exp \left(\int M(x) dx \right) (M(x)y + y') = \exp \left(\int M(x) dx \right) (y' + M(x)y) \\ &\implies \frac{d}{dx} \exp \left(\int M(x) dx \right) y = \exp \left(\int M(x) dx \right) (y' + M(x)y) \end{aligned}$$

We have done that because in the parenthesis on the left popped up the left-hand part of the original ODE.

Let us go back and multiply both side by the exponential function of the integral.

$$\exp \left(\int M(x) dx \right) (y' + M(x)y) = \exp \left(\int M(x) dx \right) N(x)$$

On the left hand part we can substitute it by derivative we have just done.

$$\frac{d}{dx} \exp \left(\int M(x) dx \right) y = \exp \left(\int M(x) dx \right) N(x)$$

We now multiply both side by dx and take the integral of both parts.

$$\begin{aligned} d \left[\exp \left(\int M(x) dx \right) y \right] &= \exp \left(\int M(x) dx \right) N(x) dx \\ \int d \left[\exp \left(\int M(x) dx \right) y \right] &= \int \exp \left(\int M(x) dx \right) N(x) dx \end{aligned}$$

On the right-hand part very little we can do, although on the other hand we have the integral of a differential on the right hand part, and the integral of the differential of x is just x . Thus:

$$\exp \left(\int M(x) dx \right) y = \int \exp \left(\int M(x) dx \right) N(x) dx$$

To obtain the value of y we just now require to divide both side by the exponential function and we have it finally:

$$y = \frac{\int N(x) \exp \left(\int M(x) dx \right) dx}{\exp \left(\int M(x) dx \right)}$$

2.4 Bernoulli's ODE

Bernoulli's ODE are non-linear ODE which have the following form:

$$y' + M(x)y = N(x)y^n, \quad n \in \mathbb{R} \setminus \{-1, 0\}$$

This ODE is non-linear because of the y^n term, but if the equation satisfy that form, we can make a trick to turn this into linear and have a general formula for this.

We start by multiplying both side by y^{-n} , distribute it and use the power's rules.

$$\begin{aligned} y^{-n}(y' + M(x)y) &= y^{-n}N(x)y^n \\ \implies y^{-n}y' + y^{-n}M(x)y &= y^{-n}N(x)y^n \\ \implies y^{-n}y' + M(x)y^{-n+1} &= N(x)y^{-n+1} \\ \implies y^{-n}y' + M(x)y^{1-n} &= N(x)y^{1-n} \end{aligned}$$

Now we need to make a change of variable and define $z = y^{1-n}$ and take its derivative:

$$\frac{dz}{dx} = (1-n)y^{-n}y' \implies \frac{1}{1-n} \frac{dz}{dx} = y^{-n}y'$$

We see that the result of differentiating z is also in the equation. We can rewrite the whole equation in terms of z like:

$$\frac{1}{1-n} \frac{dz}{dx} + M(x)z = N(x)z$$

We multiply both side by $1-n$.

$$\frac{dz}{dx} + (1-n)M(x)z = (1-n)N(x)z$$

This result is a linear non-homogeneous ODE, allowing us to solve this equation using the general formula we have just derived in the previous chapters.

$$\begin{aligned}
z &= \frac{\int (1-n)N(x) \exp\left(\int (1-n)M(x)dx\right) dx}{\exp\left(\int (1-n)M(x)dx\right)} \\
&= \frac{(1-n) \int N(x) \exp\left((1-n) \int M(x)dx\right) dx}{\exp\left((1-n) \int M(x)dx\right)} \\
&= \frac{(1-n) \int N(x) \exp(1-n) \exp\left(\int M(x)dx\right) dx}{\exp(1-n) \exp\left(\int M(x)dx\right)} \\
&= \frac{(1-n) \exp(1-n) \int N(x) \exp\left(\int M(x)dx\right) dx}{\exp(1-n) \exp\left(\int M(x)dx\right)} \\
&= \frac{(1-n) \int N(x) \exp\left(\int M(x)dx\right) dx}{\exp\left(\int M(x)dx\right)} \\
&= \frac{1}{\exp\left(\int M(x)dx\right)} (1-n) \int N(x) \exp\left(\int M(x)dx\right) dx
\end{aligned}$$

We cannot forget we just got the value for z but we are actually interested in knowing y . Hence, since $z = y^{1-n}$, $y = z^{\frac{1}{1-n}}$.

$$\therefore y = \left[\frac{1}{\exp\left(\int M(x)dx\right)} (1-n) \int N(x) \exp\left(\int M(x)dx\right) dx \right]^{\frac{1}{1-n}}$$

3 Laplace's transform

The Laplace's transform $\mathcal{L}$ over a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined as:

$$\mathcal{L}\{f(t)\} := \int_0^\infty f(t)e^{-st}dt$$

This operator is super useful in differential equations because it converts an ordinary differential equation into an algebraic equation and a partial differential equation to an ordinary differential equation.

3.1 Laplace's transform over ordinary derivatives

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function in the interval $[a, b]$ and differentiable in the interval (a, b) . Thus:

$$\mathcal{L}\{f^{(n)}(t)\} = \int_0^\infty f^{(n)}(t)e^{-st}dt$$

Now, we can use the method of integration by parts as we have a multiplication and one of the terms is an exponential.

$$\begin{aligned}
&= f^{(n-1)}(t)e^{-st} \Big|_{t=0}^{t=\infty} - \int_0^\infty -se^{-st} f^{(n-1)}(t) dt = f^{(n)}(t)e^{-st} \Big|_{t=0}^{t=\infty} + \int_0^\infty se^{-st} f^{(n-1)}(t) dt \\
&= f^{(n-1)}(0) + s \int_0^\infty e^{-st} f^{(n-1)}(t) dt = f^{(n-1)}(0) + s\mathcal{L}\{f^{(n-1)}(t)\}
\end{aligned}$$

Therefore:

$$\mathcal{L}\{f^{(n)}(t)\} = -f^{(n)}(0) + s\mathcal{L}\{f^{(n-1)}(t)\}$$

Nonetheless, this result is recursive, because if $n \geq 2$, we have now again the Laplace's transform of a derivative again. Hence, we can apply over and over the formula

$$\begin{aligned}
\mathcal{L}\{f^{(n)}(t)\} &= -f^{(n-1)}(0) + s\mathcal{L}\{f^{(n-1)}(t)\} \\
&= -f^{(n-1)}(0) - sf^{(n-2)}(0) + s^2\mathcal{L}\{f^{(n-2)}(t)\} \\
&= -f^{(n-1)}(0) - sf^{(n-2)}(0) - s^2f^{(n-3)}(0) + s^3\mathcal{L}\{f^{(n-3)}(t)\} \\
&= -f^{(n-1)}(0) - sf^{(n-2)}(0) - s^2f^{(n-3)}(0) - s^3f^{(n-4)}(0) + s^4\mathcal{L}\{f^{(n-4)}(t)\} \\
&= -f^{(n-1)}(0) - sf^{(n-2)}(0) - s^2f^{(n-3)}(0) - s^3f^{(n-4)}(0) - s^4f^{(n-5)}(0) + s^5\mathcal{L}\{f^{(n-5)}(t)\} \\
&\vdots
\end{aligned}$$

$$\therefore \mathcal{L}\{f^{(n)}(t)\} = s^n \mathcal{L}\{f(t)\} - \sum_{\mu=0}^{n-1} s^\mu f^{(n-\mu-1)}(0)$$

We see how the Laplace's transform took away all the ordinary derivatives it contained, gaining a lot of potential when working with differential equations.

3.2 Linearity

Let $f_\mu : \mathbb{R} \rightarrow \mathbb{R}$ and $\sigma_\mu \in \mathbb{F}$:

$$\begin{aligned}
\mathcal{L}\left\{\sum_{\mu=0}^n \sigma_\mu f_\mu(t)\right\} &= \int_0^\infty \sum_{\mu=0}^n \sigma_\mu f_\mu(t) e^{-st} dt = \sum_{\mu=0}^n \int_0^\infty \sigma_\mu f_\mu(t) e^{-st} dt = \sum_{\mu=0}^n \sigma_\mu \int_0^\infty f_\mu(t) e^{-st} dt \\
&= \sum_{\mu=0}^n \sigma_\mu \mathcal{L}\{f_\mu(t)\}
\end{aligned}$$